Summary of activities, 16 March 2026

1. Reprogramming PD07 fibroblasts to iPSCs

I revised Jinny's existing protocol to begin reprogramming the PD07 fibroblasts (P6) to iPSCs, starting Monday (16Mar26). To begin, I prepared NM-RNA reprogramming cocktail and aliquoted it out (in 8.5-μL volumes; each well takes 7.7 μL) and stored in a box with my name on it in the -80°C. I also prepared an aliquot of the remaining mir-200c solution in nuclease-free water (in a 2.2-μL volume; although each well only requires 0.25 μL, Bin informed me that the mir-200c is not expensive, and accuracy took priority over waste) and stored in the same box in the -80°C. Over days 1–4, I ensured that media was changed in the morning (at 10:00AM) and transfection was initiated overnight (at 4:00PM), as directed. This is all detailed on my LabArchives.

2. PD07 fibroblast quality control

In line with Figure 5 of the INTERACT package, I began quality control testing for the PD07 fibroblasts. The line is free from mycoplasma, bacterial, and endotoxin contamination (see attached). I believe we can also assemble a figure of representative images of the PD07 fibroblasts in culture from our microscopy images, as in Figure 4 of the INTERACT package.

3. Gene editing and electroporation

I made revised worksheets (with the correct kit # and improved clarity) for GFP and gene editing electroporation protocols. I also made an explicit plan to complete establishing a strategy for gene editing in PD07 (namely, testing prime editors and mRNA delivery format); these documents are all attached here. Specific questions regarding Kiara's data and how to best proceed are indicated in red in the *gene editing planning* document. I'll enumerate them here as well, for convenience:

1. Are the "+21G" and "19mer" experiments that Kiara conducted different strategies to evade MMR (i.e., by installing concomitant silent edits near the targeted mutation site)?

2. How should I interpret the last slide from Kiara's presentation, which suggests that the conclusions from the HEK293T cells do not necessarily translate into primary cells? How should this inform our experimental planning?
3. Does it make more sense to test our intended conditions (prime editor, mRNA delivery format) sequentially (i.e., determine the most effective prime editor first, then the best mRNA delivery format) or combinatorially (i.e., test all possible combinations of PE+mRNA simultaneously)?

4. PD07 *GBA1* correction *in vitro*

As a follow-up to my work designing a battery of tests to demonstrate that *GBA1* correction recovers functional deficits, I have begun to create worksheets detailing the various assays. As with the above, uncertain information is indicated in red. These worksheets are a work in progress, but I have included them here. I have drafted worksheets for autophagic flux, αS (total and phosphorylated protein via Western blot), αS (with PFF treatment), and a protocol for general cell harvest for Western blotting. To do: GCase enzyme activity, lipid droplet staining (via ICC), GCase-LAMP1 co-localization (via ICC), and cell viability (via CellTiter-Glo).

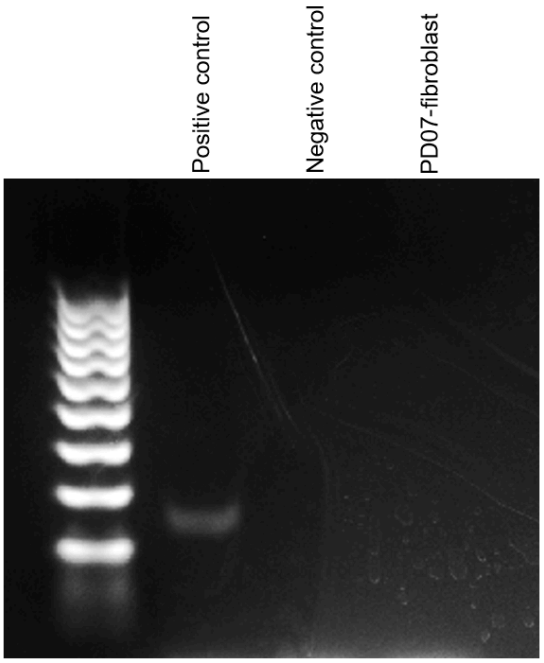
5. Miscellaneous

Pursuant to my conversation with Bin, I will be responsible for the TC room equipment maintenance — as well as just generally making sure that things are able to run smoothly in the TC, as we all need it for our research. I have created a worksheet (attached) for the weekly maintenance and monitoring of TC room equipment to facilitate this. I also worked with Kritika to begin cataloguing our reagent stocks and aliquots, especially those that are expired. We have been recording this information in our newly made cell culture reagent inventory sheet on Teams.

A.

BacT report will go here on Tuesday, 24Mar2026, after seven days.

B.



C.

```
********* ENDOSAFE Test Record *********
              PTS150 V11.1.1
DateTime: ......... 17 Mar 2026 14:14:01
Device: ........................ 23481223
Analyst: ............................ TM
Cartridge: ............... Endotoxin-LAL
Temperature: .. Start: 37.0C  End: 37.0C
Method: ......................... KX-122
Cartridge Lot#: ............. 2520301152
Cartridge Cal Code: ....... 513635262125
Range: ........................... 5-0.05
Range Time: ............... Sec: 136-752
Onset Time(s): ....... >752 290 >752 310
Endotoxin Limit: ............ 0.05 EU/mL
Slope: -0.371 ........ Intercept: +2.393
Dilution: ......................... 1:50
Sample Lot: ........................ N/A
Sample Name: ........... PD07FBSM11MAR26
Sample Rxn Time CV: .......... 0.0% Pass
Spike Value: ............... 0.544 EU/mL
Spike Rxn Time CV: ........... 4.7% Pass
Spike Recovery: ............... 133% Pass
Test Suitability: ................. Pass
Sample Endotoxin Limit: ...... Undefined
Sample Value: ............... <2.50 EU/mL
$d5e9719bdafce3a591bb233561d10cc7fd7d56af
```

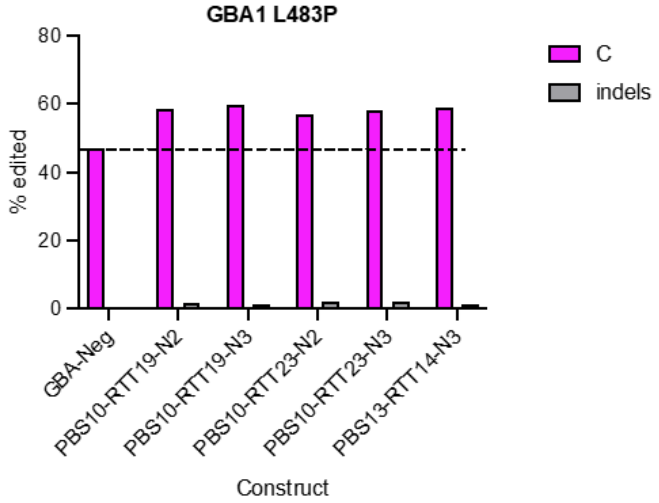
Figure 1. Quality control testing of PD07 fibroblast cultures. Tissue culture supernatant collected from PD07 fibroblasts on the day of harvest (11Mar2026) underwent testing for (A) the presence of bacteria (BacT/ALERT, Biomerieux): no bacterial contamination was detected following seven days of incubation; (B) the presence of mycoplasma via PCR: the positive control shows a clear band, while none is present for the sample; and (C) the presence of endotoxin (Endosafe, Charles River): endotoxin testing showed a sample value of <2.5 EU/mL; the max threshold pyrogenic dose is less than 0.003 U/kg/hour (2.5 U/mL x 0.0012 mL/kg/hour), which meets the QC criteria of <0.2 EU/kg/hour.

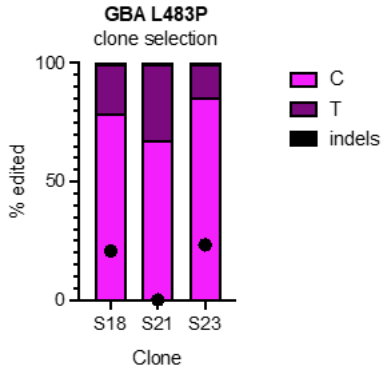
PD07 gene editing plan

Project priority: *Establish a plan to complete optimization of the gene editing strategy for PD07, in preparation for one-step strategy optimization. I have included Kiara's preliminary data here for easy reference and to check my understanding, given that I was not present for her initial presentation.*

The workflow for establishing a gene editing strategy, as laid out in the INTERACT document, is as follows: (1) a HEK293T reporter cell line expressing the mutation is established; (2) pegRNAs are screened for the highest efficiency by systematically varying the primer-binding sites (PBS) and reverse transcriptase template (RTT) lengths alongside mismatch repair (MMR) evasion strategies; (3) nicking gRNAs (ngRNAs) for PE3 systems are screened for the highest efficiency in combination with pegRNAs; (4) an evolved prime editing (PE) strategy is selected using the tested pegRNA and ngRNA; (5) finally, both linear and circular mRNA delivery formats are screened.

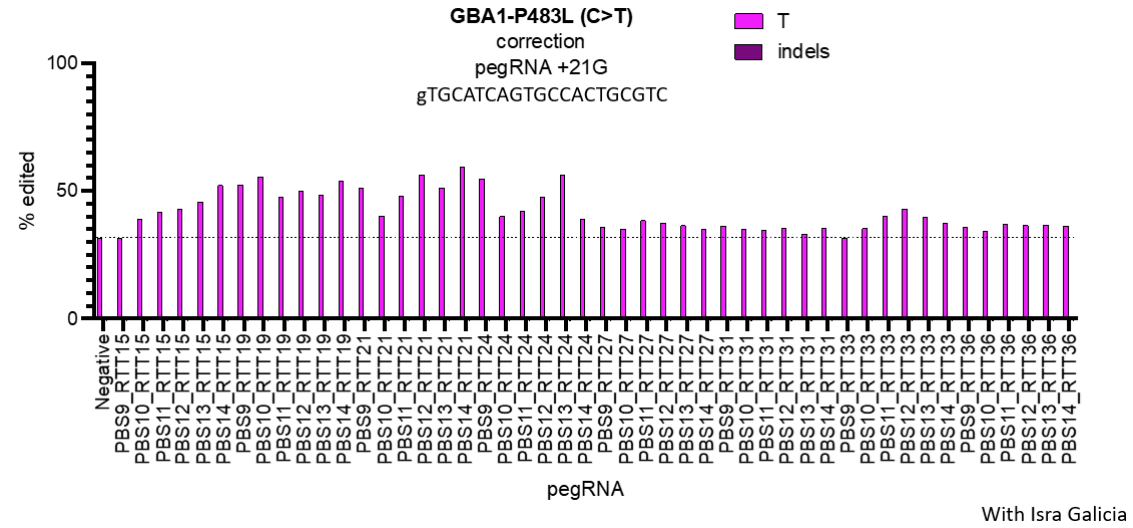
Based on Kiara's preliminary data, I believe we have completed steps 1–3. (1) The L444P HEK293T line has been established, noting that the *GBAP1* pseudogene, which naturally contains the L444P mutation, contributes to the high apparent baseline L444P content. The S21 clone was selected for the absence of indels apposed to the mutation site:



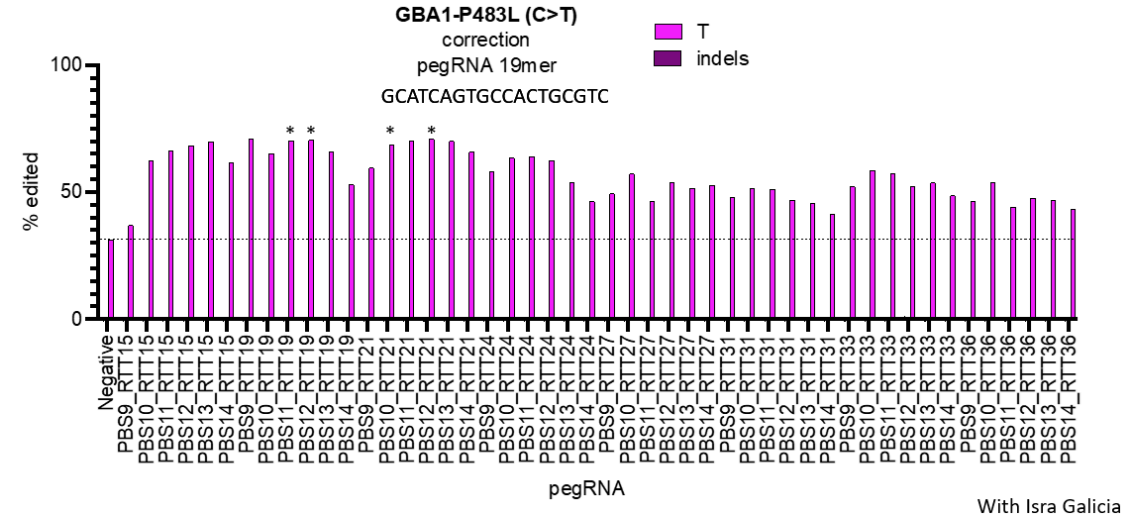


(2) pegRNAs were screened for both a +21G and 19mer strategy, identifying four candidate pegRNAs with varying PBS and RTT lengths: pegRNA222, 223, 228 and 230. My assumption here is that the +21G and 19mer approaches here refer to strategies installing concomitant silent mutations to enable evasion of MMR — please correct me if I am wrong.

A.

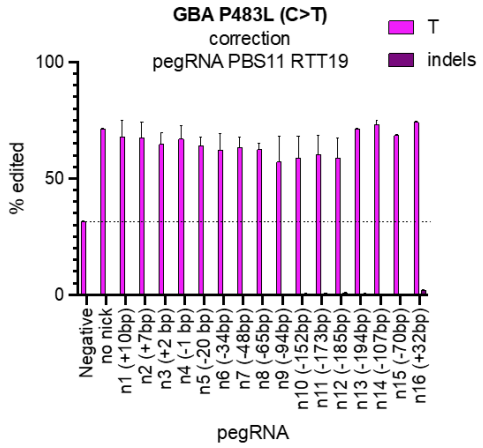


B.

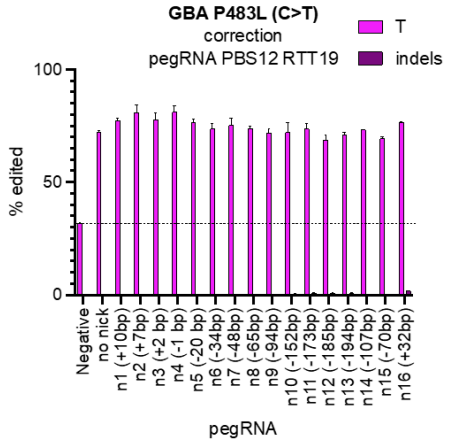


(3) These four candidate pegRNAs were then screened for ngRNA efficiency. ngRNA2 was the most efficient overall:

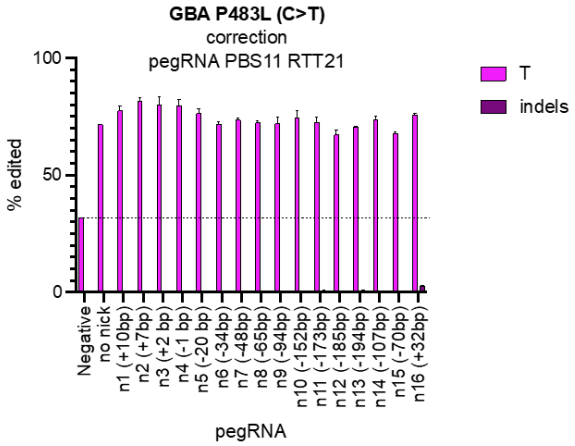
A. pegRNA222



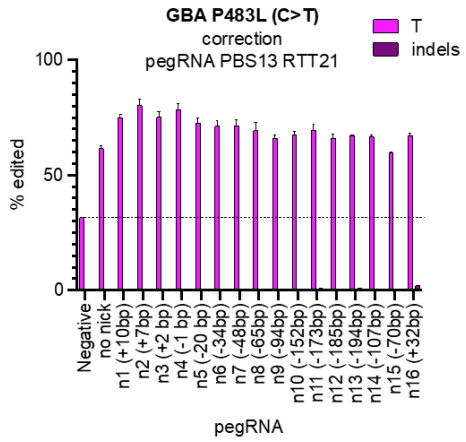
B. pegRNA223



C. pegRNA 228



D. pegRNA230



Kiara seems to have selected pegRNA223 based on these data, although it seems to me that both 228 and 230 exhibited similar if not greater editing efficiency with ngRNA2.

In summary (see the table below), we have selected our pegRNA and ngRNA based on the HEK293T reporter line data. What remains are steps 4 and 5: to identify our most efficient prime editing system (between PEMax and PE6d), and to determine whether the linear or circular mRNA delivery format is more efficient. One of my concerns regards Kiara's last slide, which suggests that these conclusions have not necessarily translated to efficient gene editing in primary cells. If we cannot establish effective editing in our PD07 fibroblasts, it may make sense to test some of the other competitive pegRNA and ngRNA candidates from the HEK293T data as well.

pegRNA	**ngRNA**	**Prime editor**	**mRNA format**
223	2	PEMax	Linear
			Circular
		PE6d	Linear
			Circular

I suspect that the easiest way to test these is all at once — that is, rather than screening for a prime editing system first (PEMax vs. PE6d), we should simultaneously test mRNA format (as above). One question is whether there is an obvious disparity in the cost/ease-of-preparation between prime editors and mRNA formats; if one is more expensive or takes much longer to prepare, it may make sense to test these individually. Otherwise, this would involve testing a total of <u>four</u> substrate groups, run in triplicate, for a total of 12 reactions requiring 2.6×10^6 fibroblasts, per the electroporation worksheet.

I can easily maintain cultures of PD07-FB from my practice-run with the GFP electroporation. The remaining step would be to ensure we have all of the required reagents in advance. Kritika has shown me where the pegRNA223 and ngRNA2 are (in the −80°C in a box labeled "GE" for gene editing); I will try to determine what amount these vials contain without thawing them. If we need any other reagents (e.g., PEMax, PE6d), I will establish a plan with Bin to either order these or consult our collaborators in the Liu Lab.

Massachusetts General Hospital
Founding Member, Mass General Brigham

George A. "Doc" Lopez, MD
Laboratory for Regenerative Cell Therapy
White 536, 55 Fruit St, Boston, MA 02114

Process	**α-synuclein aggregation with preformed fibrils (PFFs) *in vitro***		
Date & time		Operator	

Bill of materials:

Cells	Midbrain dopaminergic progenitors	ID	
Passage #		Plate size	
Day		# plates (wells)	
Density		**Scribe**	

Reagent	Lot #	Quantity/unit	# units	Expiry	**Scribe**
Media					
PFFs					
TX-100					
Tris-buffered saline					
Protease inhibitors					
Phosphatase inhibitors					
SDS (10%)					
DBPS					

Days 21–27:

Instruction	**Data**	**Scribe**
On D21, treat DAPs with 0.5 μM PFFs in 2 mL of media. Change media (with no PFFs added) on D23, D25, and D27.	D21 with PFFs: □ D23: □ D25: □ D27: □	

Harvest:

Instruction	**Data**	**Scribe**
Cool the centrifuge to 4°C in advance and prepare a bucket of (wet) ice.	□	
On D28, harvest via differential detergent extraction. First, prepare your reagents:	□ □	

Soluble fraction (1% TBS)		
TBS	10 mL	
TX-100	100 μL	
Protease inhibitor	1:1000	
Phosphatase inhibitor	1:100	
Total	10 mL	

Insoluble fraction (2% SDS)		
TBS	800 μL	
SDS (10%)	200 μL	
Protease inhibitor	1:1000	
Phosphatase inhibitor	1:100	
Total	1 mL	

Vortex both to mix. Keep the soluble fraction solution on ice and the insoluble fraction solution at room temperature (to prevent the precipitation of the SDS).

Prepare a 1.5-mL tube per well in the BSC to collect your cells after harvest.	□	
Remove your cells from the incubator. Aspirate the media and wash with DPBS.	□	

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Aspirate, then add another 1 mL of DPBS and use a cell scraper to harvest the cells. Collect each well in your tubes.	◻	
Centrifuge at 5.0 rcf for 3 minutes at 4°C.	◻ ___ rcf ___ minutes	
Carefully aspirate the supernatant. Resuspend the pellet with 200 µL of your soluble fraction solution per well, pipetting up and down gently to mix. Incubate on ice for 30 minutes.	◻ Time: _____	
Centrifuge at 13.0 rcf for 40 minutes at 4°C. During this time, prepare a second set of 1.5-mL tubes to collect the soluble fraction.	◻ ___ rcf ___ minutes	
Using a pipet, reserve the supernatant(s) in your second (set of) tube(s). This is the soluble fraction of your harvest.	◻	
Resuspend the pellet with 200 µL of your insoluble fraction solution per well, pipetting up and down gently to mix.	◻	
Sonicate 10 times at a 1-sec. pulse on 90% power to completely disperse the pellet.	◻ _____ power	
Centrifuge at 13.0 rcf for 40 minutes at 4°C. During this time, prepare a third set of 1.5-mL tubes to collect the soluble fraction.	◻ ___ rcf ___ minutes	
Using a pipet, reserve the supernatant(s) in your third (set of) tube(s). This is the insoluble fraction of your harvest.	◻	
NOTE: This harvest can be completed without the PFF seeding to look at total and phosphorylated αS as well.		

All steps completed? ◻

Scribe: _____
(Initial & date)

Notes:

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Process	**Cell harvest for Western blot**		
Date & time		Operator	

Bill of materials:

Cells	Midbrain dopaminergic progenitors	ID	
Passage #		Plate size	
Day		# plates (wells)	
Density		**Scribe**	

Reagent	Lot #	Quantity/unit	# units	Expiry	**Scribe**
NP-40 lysis buffer (BP-119)					
DPBS					
Protease inhibitor					
Phosphatase inhibitor					

Harvest:

Instruction	**Data**	**Scribe**
Cool the centrifuge to 4°C in advance and prepare a bucket of (wet) ice.	◻	
Prepare a 1.5-mL tube per well in the BSC to collect your cells after harvest.	◻	
Remove your cells from the incubator. Aspirate the media and wash with DPBS.	◻	
Aspirate, then add another 1 mL of DPBS and use a cell scraper to harvest the cells. Collect each well in your tubes.	◻	
Centrifuge at 5.0 rcf for 3 minutes at 4°C.	◻ ___ rcf ___ minutes	
During this time, prepare your lysis buffer, to be kept on ice:	◻	
Carefully aspirate the supernatant. Resuspend the pellet with 50 µL of lysis buffer, then incubate for 15 minutes on ice.	◻ Time: _____	
Centrifuge at max. speed for 30 minutes at 4°C.		
Using a pipet, reserve the supernatant as your lysate. Discard the pellets.		

Lysis buffer	Volume added:
NP-40 lysis buffer	
Protease inhibitor (1:1000)	
Phosphatase inhibitor (1:100)	
Total	

All steps completed? ◻

Scribe: _____
(Initial & date)

Notes:

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Process	**α-synuclein Western blot**		
Date & time		Operator	

Bill of materials:

Lysates		ID	
Harvest date		# samples	
Harvest type		**Scribe**	

Reagent	Lot #	Quantity/unit	# units	Expiry	**Scribe**
Formaldehyde					
LDS (5x)					
MES buffer					
Ladder					

Running the gel:

Instruction	**Data**	**Scribe**
Prepare enough 1.5-mL tubes to accommodate your samples.	◻	
Add 12 µL per sample and 3 µL 5x LDS per tube.	◻	
Boil samples for 4 minutes in the hot rack, then briefly spin down to collect and mix.	◻	
Prepare your 1x MES running buffer.	◻	
Load 3 µL of ladder into the first lane, then 15 µL of your samples per well.	◻	
Run at 50V to begin, then proceed to 80–85V.	◻	

Transferring and imaging:

Instruction	**Data**	**Scribe**
Transfer the gel.	◻	
Prepare 4% paraformaldehyde to fix the blot. Incubate for 4 minutes with agitation.	◻	
Block for 45 minutes in 5% milk TBS with agitation.	Time in: _____	
During this time, prepare your primary antibodies by diluting in 5% milk TBS at 1:1000. Incubate for 1 hour at room temperature or overnight at 4°C.	Temp.: _____ Time in: _____	
Wash for 10 minutes in TBS-T three times with agitation.	◻ ◻ ◻	
During this time, prepare your secondary antibodies by diluting in in 5% milk TBS at 1:1000. Incubate for 30 minutes at room temperature, then wash as before and image accordingly.	Temp.: _____ Time in: _____ ◻ ◻ ◻	

All steps completed? ◻

Scribe: _____
(Initial & date)

Notes:

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Process	**Lysosomal-autophagic flux**		
Date & time		Operator	

Bill of materials:

Cells	Midbrain dopaminergic progenitors	ID	
Passage #		Plate size	
Day		# plates (wells)	
Density		**Scribe**	

Reagent	Lot #	Quantity/unit	# units	Expiry	**Scribe**
Media					
Bafilomycin A1					
Rapamycin					
Anti-LC3 primary					
Anti-actin primary					
Anti-p62 primary					

Days 21:

Instruction	**Data**	**Scribe**
4 hours prior to harvest, treat the cells with autophagy factors to stimulate (rapamycin) or inhibit (bafilomycin A1) autophagy: First, prepare your factors by diluting in media: **Bafilomycin A1 media**: Media 10 mL ☐; Bafilomycin A1 200 nM ☐ **Rapamycin media**: Media 10 mL ☐; Rapamycin 20 nM ☐ Untreated wells should receive media with a comparable dose of DMSO.	☐	
Remove your cells from the incubator. Aspirate the media and add 1 mL treatment media per well. Return the cells to the incubator for 4 hours. Note which treatments are added to each well.	☐ Time in: _____	

Harvest:

Instruction	**Data**	**Scribe**
4 hours after treatment, harvest cells as described in the **cell harvest for Western blot** worksheet.	☐	
Blot for LC3, p62, and actin. Lysosomal flux can be measured by taking the difference between		

All steps completed? ☐

Scribe: _____
(Initial & date)

Notes:

MWKST-TC-maintenance-HH

Massachusetts General Hospital Founding Member, Mass General Brigham	George A. "Doc" Lopez, MD Laboratory for Regenerative Cell Therapy White 536, 55 Fruit St, Boston, MA 02114

Process	**TC equipment maintenance**		
Date & time	20Mar26 3:00PM	Operator	HHsiang

Tasks:

Instructions	**Data**	**Scribe**
Prepare fresh 70% isopropyl alcohol	☑ 18Mar26	HH
Clean:	☑	HH
Surfaces	☑	HH
Shelves	☑	HH
Doors	☑	HH
Chairs	☑	HH
Cart (including wheels)	☑	HH
Plastic tubs (beneath vacuums)	☑	HH
Floor	☑	HH
Take out the trash:	☑	HH
Regular	☑ N/A	HH
Bio-hazard	☑	HH
Replace sharps containers: spent sharps containers go in the bio-hazard trash; fresh sharps containers can be found beneath the sink	☑ Number remaining? 2 Request more? N	HH
Empty vacuums	☑ Not needed	HH
Replace consumables	☑	HH
Charge pipettors	☑	HH
Incubator:	CO_2: 5; 5.1 Temp.: 37, 37	HH
Last sterilization?	Top: ? Bottom: ?	HH
Replace water	☑ Bottom	HH
Maintenance?	Needed: ☐ N/A: ☑	HH
Fridge/freezer maintenance?	Needed: ☐ N/A: ☑	HH
BSC recertification?	Needed: ☐ N/A: ☑	HH
Microscope maintenance?	Needed: ☐ N/A: ☑ Memory: 5GB free	HH
Centrifuge maintenance?	Needed: ☐ N/A: ☑	HH
Pipette calibration?	Needed: ☐ N/A: ☑	HH
Dewar: replace liquid nitrogen?	Needed: ☐ N/A: ☑	HH

All steps completed? ☑

Scribe: HH 20Mar26
(Initial & date)

Notes: Replaced: 10mL pipets; 1.5-mL Eppendorfs; 12-well plates; 6-cm. dishes
Incubator: PLEASE RUN STERI-CYCLE; PLEASE REPLACE HEPA FILTER